

GUERNSEY, Guernsey

WMO: 038940

Lat: 49.433N

Lon: 2.598W

Elev: 102

StdP: 100.10

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.8	1.9	-4.6	2.6	2.4	-3.2	2.9	3.1	16.8	8.0	14.8	7.9	6.3	90	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	23.6	17.9	21.7	17.2	20.3	16.6	18.7	21.8	18.0	20.4	17.3	19.3	4.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.8	19.6	17.0	12.3	18.7	16.5	11.9	18.2	53.4	22.0	51.0	20.5	49.0	19.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.3	11.1	-0.7	27.4	1.7	2.7	-1.9	29.4	-2.8	31.0	-3.8	32.5	-5.0	34.5		
(5)			-2.1	20.5	1.5	1.2	-3.1	21.3	-4.0	22.0	-4.8	22.7	-5.9	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	7.2	7.0	8.0	9.5	12.1	14.7	16.7	17.0	15.6	13.4	10.3	8.2		
	DBStd	4.19	2.52	2.35	2.13	2.21	2.25	2.07	1.90	1.74	1.72	2.05	2.19	2.57		
	HDD10.0	368	88	86	67	32	4	0	0	0	0	2	22	66		
	HDD18.3	2453	344	318	321	263	192	112	59	51	83	153	240	315		
	CDD10.0	977	2	1	4	19	71	141	207	216	169	107	32	9		
	CDD18.3	21	0	0	0	0	1	3	7	8	2	0	0	0		
	CDH23.3	67	0	0	0	0	1	13	26	26	2	0	0	0		
	CDH26.7	8	0	0	0	0	0	1	3	4	0	0	0	0		
(14) Wind	WSAvg	5.7	6.9	6.4	5.9	5.3	5.3	5.0	4.9	4.7	5.2	5.8	6.4	6.8		
(15) Precipitation	PrecAvg	839	95	76	57	53	50	46	41	55	53	92	110	111		
	PrecMax	1032	206	150	171	127	119	113	83	120	139	170	229	245		
	PrecMin	626	13	14	6	0	8	5	7	10	6	18	30	25		
	PrecStd	116	48	39	33	33	30	26	20	30	30	39	49	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.4	14.8	19.0	22.2	25.0	26.2	26.3	23.1	19.4	15.7	13.5		
		MCWB	11.5	10.2	11.2	13.4	16.0	18.5	18.8	19.4	17.6	16.4	14.4	12.5		
	2%	DB	11.6	11.2	12.7	15.9	19.3	21.9	23.2	23.1	20.8	17.7	14.6	12.9		
		MCWB	10.9	10.0	10.2	12.0	14.8	16.8	17.7	18.3	17.2	15.7	13.5	12.2		
	5%	DB	11.1	10.6	11.5	13.9	17.2	19.8	21.5	21.0	19.2	16.8	14.0	12.3		
		MCWB	10.3	9.6	9.7	11.0	13.6	15.9	17.0	17.3	16.4	15.2	12.9	11.4		
	10%	DB	10.5	10.0	10.7	12.6	15.5	18.1	20.0	19.6	18.1	16.1	13.4	11.7		
		MCWB	9.7	9.1	9.3	10.3	12.9	14.9	16.6	16.7	15.7	14.7	12.2	10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.2	11.6	14.5	17.1	19.4	19.5	20.2	18.7	17.1	15.0	13.1		
		MCDB	12.0	11.5	13.7	17.8	20.6	24.0	23.9	25.0	21.4	18.4	15.2	13.2		
	2%	WB	11.2	10.5	10.8	12.4	15.3	17.5	18.4	18.8	17.7	16.3	13.9	12.4		
		MCDB	11.4	10.9	11.9	14.9	18.1	20.9	21.8	21.6	19.6	17.2	14.3	12.7		
	5%	WB	10.5	9.9	10.2	11.5	14.2	16.3	17.6	18.0	17.1	15.7	13.2	11.7		
		MCDB	10.9	10.3	11.0	13.3	16.5	18.9	20.3	20.0	18.5	16.5	13.8	12.1		
	10%	WB	9.9	9.3	9.8	10.7	13.2	15.4	17.0	17.3	16.4	15.1	12.4	11.0		
		MCDB	10.4	9.8	10.5	12.1	15.0	17.5	19.2	18.9	17.7	15.9	13.2	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.1	3.4	3.9	4.8	5.1	5.4	5.4	5.0	4.5	3.7	3.1	3.2		
		MCDBR	3.1	3.6	4.8	6.8	7.7	8.3	8.0	7.3	5.8	4.0	3.3	3.1		
	5% WB	MCWBR	3.6	3.4	3.5	4.3	4.6	4.8	4.2	3.9	3.3	3.0	3.5	3.7		
		MCWBR	3.1	3.4	4.2	5.8	6.9	7.2	6.9	6.0	5.0	3.6	3.1	3.1		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.349	0.380	0.391	0.396	0.389	0.385	0.378	0.368	0.360	0.335	0.327		
		taud	2.411	2.406	2.333	2.330	2.371	2.418	2.430	2.472	2.477	2.474	2.450	2.376		
	Ebn at Noon		710	792	824	855	865	871	869	860	832	771	704	657		
		Edh at Noon	66	83	104	116	116	112	109	100	90	77	63	60		
(44) All-Sky Solar Radiation	RadAvg		0.87	1.64	2.87	4.65	5.61	6.10	5.90	4.93	3.76	2.12	1.09	0.71		
	RadStd		0.07	0.20	0.30	0.38	0.41	0.60	0.31	0.38	0.26	0.15	0.10	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page